

Chapter 4

Permutations and Combinations

Only One Option Correct Type Questions

- One Indian and four American men and their wives are to be seated randomly around a circular table. Then the conditional probability that the Indian man is seated adjacent to the wife given that each American man is seated adjacent to his wife is [IIT-JEE-2007 (Paper-1)]
(A) $\frac{1}{2}$ (B) $\frac{1}{3}$
(C) $\frac{2}{5}$ (D) $\frac{1}{5}$
- The letters of the word **COCHIN** are permuted and all the permutations are arranged in an alphabetical order as in a English dictionary. The number of words that appear before the word **COCHIN** is [IIT-JEE-2007 (Paper-2)]
(A) 360 (B) 192
(C) 96 (D) 48
- The number of seven digit integers, with sum of the digits equal to 10 and formed by using the digits 1, 2, and 3 only, is [IIT-JEE-2009 (Paper-1)]
(A) 55 (B) 66
(C) 77 (D) 88
- The total number of ways in which 5 balls of different colours can be distributed among 3 persons so that each person gets at least one ball is [IIT-JEE-2012 (Paper-1)]
(A) 75 (B) 150
(C) 210 (D) 243
- Six cards and six envelopes are numbered 1, 2, 3, 4, 5, 6 and cards are to be placed in envelopes so that each envelope contains exactly one card and no card is placed in the envelope bearing the same number and moreover the card numbered 1 is always placed in envelope numbered 2. Then the number of ways it can be done is [JEE (Adv)-2014 (Paper-2)]
(A) 264 (B) 265
(C) 53 (D) 67
- A debate club consists of 6 girls and 4 boys. A team of 4 members is to be selected from this club including the selection of a captain (from among these 4 members) for the team. If the team has to include at most one boy, then the number of ways of selecting the team is [JEE (Adv)-2016 (Paper-1)]
(A) 380 (B) 320
(C) 260 (D) 95
- Let $S = \{1, 2, 3, \dots, 9\}$. For $k = 1, 2, \dots, 5$, let N_k be the number of subsets of S , each containing five elements out of which exactly k are odd. Then $N_1 + N_2 + N_3 + N_4 + N_5 =$ [JEE (Adv)-2017 (Paper-2)]
(A) 125 (B) 210
(C) 252 (D) 126
- Consider 4 boxes, where each box contains 3 red balls and 2 blue balls. Assume that all 20 balls are distinct. In how many different ways can 10 balls be chosen from these 4 boxes so that from each box at least one red ball and one blue ball are chosen? [JEE (Adv)-2022 (Paper-2)]
(A) 21816 (B) 85536
(C) 12096 (D) 156816

One or More Option(s) Correct Type Questions

9. Let $S_1 = \{(i, j, k) : i, j, k \in \{1, 2, \dots, 10\}\}$,
 $S_2 = \{(i, j) : 1 \leq i < j + 2 \leq 10, i, j \in \{1, 2, \dots, 10\}\}$,
 $S_3 = \{(i, j, k, l) : 1 \leq i < j < k < l, i, j, k, l \in \{1, 2, \dots, 10\}\}$,
 and $S_4 = \{(i, j, k, l) : i, j, k \text{ and } l \text{ are distinct elements in } \{1, 2, \dots, 10\}\}$.

If the total number of elements in the set S_r is n_r , $r = 1, 2, 3, 4$, then which of the following statements is (are) TRUE?

[JEE (Adv)-2021 (Paper-2)]

- (A) $n_1 = 1000$ (B) $n_2 = 44$
 (C) $n_3 = 220$ (D) $\frac{n_4}{12} = 420$

Linked Comprehension Type Questions

Paragraph for Q.Nos. 9 and 10

Let a_n denote the number of all n -digit positive integers formed by the digits 0, 1 or both such that no consecutive digits in them are 0. Let b_n = the number of such n -digit integers ending with digit 1 and c_n = the number of such n -digit integers ending with digit 0.

[IIT-JEE-2012 (Paper-2)]

Choose the correct answer :

10. Which of the following is correct?
 (A) $a_{17} = a_{16} + a_{15}$ (B) $c_{17} \neq c_{16} + c_{15}$
 (C) $b_{17} \neq b_{16} + c_{16}$ (D) $a_{17} = c_{17} + b_{16}$
11. The value of b_6 is
 (A) 7 (B) 8
 (C) 9 (D) 11

Matrix-Match / Matrix Based Type Questions

12. Consider all possible permutations of the letters of the word ENDEANOEL. Match the Statements/Expressions in **Column-I** with the Statements/Expressions in **Column-II** and indicate your answer by darkening the appropriate bubbles in the 4×4 matrix given in the ORS.

[IIT-JEE-2008 (Paper-2)]

Column-I	Column-II
(A) The number of permutations containing the word ENDEA is	(p) $5!$
(B) The number of permutations in which the letter E occurs in the first and the last positions is	(q) $2 \times 5!$
(C) The number of permutations in which none of the letters D, L, N occurs in the last five positions is	(r) $7 \times 5!$
(D) The number of permutations in which the letters A, E, O occur only in odd positions is	(s) $21 \times 5!$

13. In a high school, a committee has to be formed from a group of 6 boys $M_1, M_2, M_3, M_4, M_5, M_6$ and 5 girls G_1, G_2, G_3, G_4, G_5 .
[JEE (Adv)-2018 (Paper-2)]

- (i) Let α_1 be the total number of ways in which the committee can be formed such that the committee has 5 members, having exactly 3 boys and 2 girls.
- (ii) Let α_2 be the total number of ways in which the committee can be formed such that the committee has at least 2 members, and having an equal number of boys and girls.
- (iii) Let α_3 be the total number of ways in which the committee can be formed such that the committee has 5 members, at least 2 of them being girls.
- (iv) Let α_4 be the total number of ways in which the committee can be formed such that the committee has 4 members, having at least 2 girls and such that both M_1 and G_1 are NOT in the committee together.

LIST-I**LIST-II**

- P. The value of α_1 is
- Q. The value of α_2 is
- R. The value of α_3 is
- S. The value of α_4 is

1. 136
2. 189
3. 192
4. 200
5. 381
6. 461

The correct option is

(A) $P \rightarrow 4; Q \rightarrow 6; R \rightarrow 2; S \rightarrow 1$

(B) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 3$

(C) $P \rightarrow 4; Q \rightarrow 6; R \rightarrow 5; S \rightarrow 2$

(D) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 1$

Integer / Numerical Value Type Questions

14. Let $n_1 < n_2 < n_3 < n_4 < n_5$ be positive integers such that $n_1 + n_2 + n_3 + n_4 + n_5 = 20$. Then the number of such distinct arrangements $(n_1, n_2, n_3, n_4, n_5)$ is
[JEE (Adv)-2014 (Paper-1)]
15. Let $n \geq 2$ be an integer. Take n distinct points on a circle and join each pair of points by a line segment. Colour the line segment joining every pair of adjacent points by blue and the rest by red. If the number of red and blue line segments are equal, then the value of n is
[JEE (Adv)-2014 (Paper-1)]
16. Let n be the number of ways in which 5 boys and 5 girls can stand in a queue in such a way that all the girls stand consecutively in the queue. Let m be the number of ways in which 5 boys and 5 girls can stand in a queue in such a way that exactly four girls stand consecutively in the queue. Then the value of $\frac{m}{n}$ is
[JEE (Adv)-2015 (Paper-1)]
17. Words of length 10 are formed using the letters $A, B, C, D, E, F, G, H, I, J$. Let x be the number of such words where no letter is repeated; and let y be the number of such words where exactly one letter is repeated twice and no other letter is repeated. Then, $\frac{y}{9x} =$
[JEE (Adv)-2017 (Paper-1)]
18. The number of 5 digit numbers which are divisible by 4, with digits from the set $\{1, 2, 3, 4, 5\}$ and the repetition of digits is allowed, is _____.
[JEE (Adv)-2018 (Paper-1)]

19. Five persons A, B, C, D and E are seated in a circular arrangement. If each of them is given a hat of one of the three colours red, blue and green, then the number of ways of distributing the hats such that the persons seated in adjacent seats get different coloured hats is _____. **[JEE (Adv)-2019 (Paper-2)]**
20. An engineer is required to visit a factory for exactly four days during the first 15 days of every month and it is mandatory that no two visits take place on consecutive days. Then the number of all possible ways in which such visits to the factory can be made by the engineer during 1-15 June 2021 is _____. **[JEE (Adv)-2020 (Paper-2)]**
21. In a hotel, four rooms are available. Six persons are to be accommodated in these four rooms in such a way that each of these rooms contains at least one person and at most two persons. Then the number of all possible ways in which this can be done is _____. **[JEE (Adv)-2020 (Paper-2)]**
22. The number of 4-digit integers in the closed interval $[2022, 4482]$ formed by using the digits 0, 2, 3, 4, 6, 7 is _____. **[JEE (Adv)-2022 (Paper-1)]**



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